10 Hard Problems

Random Walks & Gambler's Ruin

Chapter 4.3 · Jane Street National Probability Challenge · Stage 1 (HackerEarth) style

Plan of attack

Stage 1 is a timed, numeric-answer test, so each problem below has a *target time* and ends in a clean number. Try each problem for the target time with the solution covered. If you miss the time, read the solution and re-solve it from scratch the next day.

#	Problem	Chapter 4.3 skill covered	Level
1	Take-profit / stop-loss trade	Fair ruin, duration, martingale EV	★★
2	Biased trader, symmetric target	Biased ruin, halfway trick, Wald	★★★
3	How long does a winner take?	Conditioning on the exit (Doob transform)	★★★★★
4	The downhill stock and its peak	Infinite bank, all-time maximum	★★★
5	Touching a barrier in 20 steps	Reflection principle	★★★★
6	Buy/sell order flow	Ballot theorem, Catalan, first return	★★★★
7	Inventory with a wall	Reflecting barrier, unfolding	★★★★
8	The ant on a hexagon	Cycles, cover time, last vertex	★★★★
9	Asymmetric barriers, biased coin	Ruin, duration, Wald consistency	★★★
10	1D vs 2D vs 3D	Return probabilities, recurrence	★★★★

Formula sheet (what you need for all ten)

Walk ± 1 with $P(+1) = p$, $q = 1 - p$, $r = q/p$. Start k, absorbing at 0 and N.

$$W_k = \frac{k}{N} \quad (p = q), \qquad W_k = \frac{1 - r^k}{1 - r^N} \quad (p \neq q), \qquad W_k = \frac{1}{1 + r^k} \text{ if } N = 2k.$$

$$\mathbb{E}[T] = k(N - k) \quad (p = q), \qquad \mathbb{E}[T] = \frac{NW_k - k}{p - q} \quad (p \neq q) \text{ (Wald)}.$$

Infinite bank: $\mathbb{P}(\text{ever hit } 0 \mid \text{start } k) = \min\{1, (q/p)^k\}$. Reflection: $\mathbb{P}(\max_{i \le n} S_i \ge m) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)$. Ballot: strictly ahead $\frac{a-b}{a+b}$, never behind $\frac{a-b+1}{a+1}$, some tie $\frac{2b}{a+b}$. $u_{2n} = \binom{2n}{n} 4^{-n}$, $C_n = \frac{1}{n+1}\binom{2n}{n}$.

Part I: Warm-up with a twist

Problem 1 | The take-profit / stop-loss trade

Difficulty: ⋆⋆ **Target time:** 2 min **Topic:** fair ruin, duration, martingale

A stock's price moves up or down by exactly \$1 each second, each with probability $\frac{1}{2}$, independently. You buy at \$100 and will sell the first moment the price touches \$110 (take profit) or \$95 (stop loss).

(a) What is the probability you take profit?

(b) What is the expected holding time in seconds?

(c) What is your expected P&L before costs?

(d) You pay funding costs of \$0.01 per second held. What is your expected P&L after costs?

Step-by-step solution

Step 1. Shift so the stop loss is state 0: the start is $k = 5$ and the take profit is $N = 15$. The walk is fair.

Step 2. (a) $\mathbb{P}(\text{profit}) = k/N = 5/15 = \frac{1}{3}$.

Step 3. (b) $\mathbb{E}[T] = k(N-k) = 5 \cdot 10 = 50$ seconds.

Step 4. (c) Profit is +\$10 w.p. $\frac{1}{3}$ and −\$5 w.p. $\frac{2}{3}$. So $\mathbb{E}[\text{P\&L}] = \frac{10}{3} - \frac{10}{3} = 0$. This must be true: the price is a martingale, so optional stopping says the expected exit price equals the entry price.

Step 5. (d) $\mathbb{E}[\text{cost}] = 0.01 \cdot 50 = \$0.50$, so net $= -\$0.50$.

Answer. (a) $\frac{1}{3}$ (b) 50 s (c) \$0 (d) −\$0.50.

Why it works – the big idea

In a fair game the average fortune never moves, so the exit values (110 w.p. x, 95 w.p. $1-x$) must average to 100: $110x + 95(1-x) = 100 \Rightarrow x = \frac{1}{3}$. You never need a recurrence. Duration is the *area of the rectangle* formed by your distances to the two barriers: 5×10.

Analogy – make it intuitive

A seesaw with a heavy kid (the far, wide-open \$10 side) and a light kid (the near \$5 stop). The pivot stays put, so the chance of the far end winning is exactly the near-distance share $\frac{5}{15}$.

Part II: Biased games and conditioning

Problem 2 | The biased trader with a symmetric target

Difficulty: ⋆⋆⋆ **Target time:** 3 min **Topic:** biased ruin, halfway trick, Wald

Each round you win \$1 with probability $\frac{2}{3}$ and lose \$1 with probability $\frac{1}{3}$. You start with \$3 and quit at \$6 or \$0.

(a) Find $\mathbb{P}(\text{reach } \$6)$.

(b) Find the expected number of rounds.

(c) Find the probability that the game lasts *exactly* 3 rounds.

Step-by-step solution

Step 1. $r = q/p = \frac{1}{2}$, $k = 3$, $N = 6 = 2k$. Halfway trick: $W_3 = \dfrac{1}{1+r^3} = \dfrac{1}{1+1/8} = \dfrac{8}{9}$.

Step 2. (b) By Wald: $\mathbb{E}[S_T] = k + (p-q)\mathbb{E}[T]$. We have $\mathbb{E}[S_T] = NW_3 = 6 \cdot \frac{8}{9} = \frac{16}{3}$ and $p - q = \frac{1}{3}$. So

$$\mathbb{E}[T] = \frac{\frac{16}{3} - 3}{1/3} = \frac{7/3}{1/3} = 7.$$

Step 3. (c) The earliest possible finish is round 3 (distance 3). To finish in exactly 3 rounds the three moves are all up (prob. $(\frac{2}{3})^3 = \frac{8}{27}$) or all down (prob. $(\frac{1}{3})^3 = \frac{1}{27}$). Total $\frac{9}{27} = \frac{1}{3}$.

Answer. (a) $\frac{8}{9}$ (b) 7 rounds (c) $\frac{1}{3}$.

Why it works – the big idea

The two-barrier formula collapses when the start is exactly halfway, because $1 - r^{2k} = (1 - r^k)(1 + r^k)$. For the duration you never need a new recurrence: the *average gain per round* is the drift $p - q$, and the total average gain is (final wealth)−(start). Divide one by the other.

Analogy – make it intuitive

A river current pushing a boat toward the right bank. The boat starts mid-river, so a left-bank landing needs a streak of bad luck three times in a row against the current. The drift is also the boat's speed: total distance the average trip covers divided by speed gives the time.

Problem 3 | How long does a winner take? (conditioning on the exit)

Difficulty: ⋆⋆⋆⋆⋆ **Target time:** 5 min **Topic:** conditioning on the exit, Doob transform

A fair walk starts at $k = 3$ and stops at 0 or $N = 10$. Find

(a) $\mathbb{E}[T]$ unconditionally,

(b) $\mathbb{E}[T \mid \text{the walk ends at } 10]$,

(c) $\mathbb{E}[T \mid \text{the walk ends at } 0]$.

Step-by-step solution

Step 1. (a) $\mathbb{E}[T] = 3 \cdot 7 = 21$.

Step 2. Let $h_k = \mathbb{E}[T \cdot \mathbf{1}_{\text{win}}]$. Since $T = 1 + T'$ after the first step,

$$h_k = \mathbb{P}(\text{win from } k) + \tfrac{1}{2}(h_{k+1} + h_{k-1}) = \frac{k}{N} + \tfrac{1}{2}(h_{k+1} + h_{k-1}).$$

Rearranged: $h_{k+1} - 2h_k + h_{k-1} = -\dfrac{2k}{N}$, with $h_0 = h_N = 0$.

Step 3. Guess a cubic. Since $(k+1)^3 - 2k^3 + (k-1)^3 = 6k$, the function $-\dfrac{k^3}{3N}$ has second difference $-\dfrac{2k}{N}$. Add a linear term Ak to meet the boundary conditions: $h_N = 0 \Rightarrow A = \dfrac{N}{3}$. So

$$h_k = \frac{k(N^2 - k^2)}{3N}.$$

Step 4. (b) Divide by $\mathbb{P}(\text{win}) = k/N$:

$$\mathbb{E}[T \mid \text{win}] = \frac{N^2 - k^2}{3} = \frac{100 - 9}{3} = \frac{91}{3} \approx 30.33.$$

Step 5. (c) By the mirror image ($k \to N - k$): $\mathbb{E}[T \mid \text{ruin}] = \dfrac{N^2 - (N-k)^2}{3} = \dfrac{100 - 49}{3} = 17.$

Step 6. Check: $0.3 \cdot \frac{91}{3} + 0.7 \cdot 17 = 9.1 + 11.9 = 21$ ✓.

Answer. (a) 21 (b) $\frac{91}{3} \approx 30.33$ (c) 17.

Why it works – the big idea

Conditioned on winning, the fair walk is no longer fair: it becomes a walk with an upward pull. At position k it steps up with probability

$$\tfrac{1}{2} \cdot \frac{W_{k+1}}{W_k} = \frac{k+1}{2k},$$

which is large near 0 (a strong push away from the ruin edge) and tends to $\frac{1}{2}$ far from it. A conditioned walk has to hurry away from ruin, but it must also avoid hitting the goal "too easily", which is why winners are *slower* on average than the unconditioned game, and losers are faster.

Analogy – make it intuitive

Winners are survivors. Among all drunks starting at 3 on a pier of length 10, those who reach the taxi are the ones who dodge the sea over and over, bouncing around a lot first, so the winners' trips are long. The drunks who fall in the sea are the unlucky ones who fell early.

Common trap

Do not assume $\mathbb{E}[T \mid \text{win}] = \mathbb{E}[T]$. The exit is correlated with the duration. The check $\sum \mathbb{P}(\text{exit})\mathbb{E}[T \mid \text{exit}] = \mathbb{E}[T]$ is a good safety net.

Problem 4 | The downhill stock and its all-time peak

Difficulty: ⋆⋆⋆ **Target time:** 3 min **Topic:** infinite bank, maximum of a walk

A stock moves up \$1 with probability 0.4 and down \$1 with probability 0.6 each tick, forever. Starting from \$0 let M be the highest level the price *ever* reaches ($M \geq 0$).

(a) Find $\mathbb{P}(\text{ever reaches } +3)$.

(b) Find $\mathbb{E}[M]$ and $\mathrm{Var}(M)$.

(c) You place a limit sell at $+3$ and a stop at *no* lower level. What is the probability the sell order ever fills?

Step-by-step solution

Step 1. Let $\rho = p/q = \frac{0.4}{0.6} = \frac{2}{3}$. Reaching $+1$ ever is the cliff-hanger problem mirrored: probability ρ. Climbing is memoryless, so $\mathbb{P}(M \geq a) = \rho^a$.

Step 2. (a) $\mathbb{P}(M \geq 3) = \left(\frac{2}{3}\right)^3 = \frac{8}{27} \approx 0.296$.

Step 3. (b) M is geometric: $\mathbb{P}(M = a) = (1-\rho)\rho^a$, $a = 0, 1, 2, \ldots$ Hence

$$\mathbb{E}[M] = \frac{\rho}{1-\rho} = \frac{2/3}{1/3} = 2, \qquad \mathrm{Var}(M) = \frac{\rho}{(1-\rho)^2} = \frac{2/3}{1/9} = 6.$$

Step 4. (c) A limit sell at $+3$ fills iff the price ever touches $+3$, i.e. iff $M \geq 3$: $\frac{8}{27}$.

Answer. (a) $\frac{8}{27}$ (b) $\mathbb{E}[M] = 2$, $\mathrm{Var}(M) = 6$ (c) $\frac{8}{27} \approx 0.296$.

Why it works – the big idea

Each new level is a fresh "cliff-hanger": to go from a to $a+1$ you must climb one more level, with probability ρ each time, independent of the past. So the max is a geometric random variable. The same argument with $p > \frac{1}{2}$ shows $M = \infty$ with probability 1.

Analogy – make it intuitive

A mountaineer on a gently downward-sloping mountain range. Each record peak is only $\frac{2}{3}$ as likely to be reached as the previous one, so records get rarer geometrically.

Part III: Reflection, Catalan and ballots

Problem 5 | Touching a barrier in 20 steps

Difficulty: ⋆⋆⋆⋆ **Target time:** 4 min **Topic:** reflection principle

A fair ± 1 walk makes $n = 20$ steps from 0. Find

(a) $\mathbb{P}(\text{the walk touches } +4 \text{ at some time})$,

(b) $\mathbb{P}(\text{the walk touches } +4 \text{ yet ends below } 0)$,

(c) $\mathbb{P}(\max_{i \leq 20} S_i = 4 \text{ exactly})$.

(Give fractions over $2^{20} = 1048576$ and a decimal.)

Step-by-step solution

Step 1. S_{20} is even. Useful counts $\binom{20}{(20+j)/2}$: $j=4:\ 125970$; $6:\ 77520$; $8:\ 38760$; $10:\ 15504$; $12:\ 4845$; $14:\ 1140$; $16:\ 190$; $18:\ 20$; $20:\ 1$.

Step 2. (a) Reflection: $\#\{\text{touch } 4\} = 2\,\#\{S_{20} > 4\} + \#\{S_{20} = 4\}$. Now $\#\{S_{20} > 4\} = 77520 + 38760 + 15504 + 4845 + 1140 + 190 + 20 + 1 = 137980$. So

$$\# = 2(137980) + 125970 = 401930, \qquad \mathbb{P} = \frac{401930}{1048576} \approx 0.3833.$$

Step 3. (b) A path that ends at $e < 4$ and touches 4 is reflected after its first touch to a path ending at $8 - e$. For $e < 0$ the images end at $10, 12, \ldots, 20$ (with $e = -2, \ldots, -12$). Count:

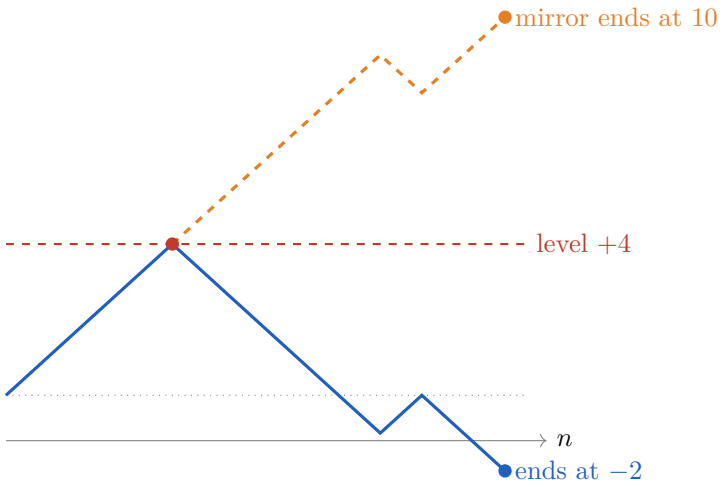
$$15504 + 4845 + 1140 + 190 + 20 + 1 = 21700, \qquad \mathbb{P} = \frac{21700}{1048576} \approx 0.0207.$$

Step 4. (c) $\mathbb{P}(\max \geq 5)$ uses the same formula with $m = 5$ (odd): $2\mathbb{P}(S > 5) + \mathbb{P}(S = 5) = 2 \cdot 137980 + 0 = 275960$ (since S_{20} is even). Therefore

$$\#\{\max = 4\} = 401930 - 275960 = 125970 = \binom{20}{12}, \qquad \mathbb{P} \approx 0.1201.$$

(In general $\mathbb{P}(\max_{i \leq n} S_i = m) = \mathbb{P}(S_n = m)$ or $\mathbb{P}(S_n = m + 1)$, whichever is possible by parity.)

Answer. (a) $\frac{401930}{2^{20}} \approx 0.3833$ (b) $\frac{21700}{2^{20}} \approx 0.0207$ (c) $\frac{125970}{2^{20}} \approx 0.1201$.



Why it works – the big idea

After the first touch of the barrier, flip every later step (up ↔ down). The new path is a valid walk that ends on the *other* side of the barrier. This operation is reversible, so "touched the barrier and ended low" is in one-to-one correspondence with "ended high". Any path ending above the barrier automatically touched it. That's why the count doubles.

Analogy – make it intuitive

A mirror placed along the barrier line. The instant a path hits the mirror it is replaced by its reflection, and the reflected twin carries on to the far side. Bad paths ↔ mirror twins.

Stage-1 speed tip

Parity first: a 20-step walk ends at an even level, so "end ≥ 5" is the same as "end ≥ 6", and $S = 5$ has probability 0. That alone avoids most off-by-one errors.

Problem 6 | Buy/sell order flow: ballot theorem, Catalan, first return

Difficulty: ⋆⋆⋆★ **Target time:** 4 min **Topic:** ballot, Catalan, first return

Six buy orders and four sell orders arrive in a uniformly random order. After each order, compare the number of buys with the number of sells so far.

(a) What is the probability that buys are *strictly ahead* after every order?

(b) What is the probability that buys are *never behind* (ties allowed)?

(c) What is the probability that there is *some tie* after the first order?

(d) A fair ± 1 walk makes 10 steps. What is the probability that its *first return to* 0 occurs exactly at step 10?

Step-by-step solution

Step 1. $a = 6$, $b = 4$, total orderings $\binom{10}{4} = 210$.

Step 2. (a) Ballot theorem: $\dfrac{a-b}{a+b} = \dfrac{2}{10} = \dfrac{1}{5}$.

Step 3. (b) Orderings that go behind touch -1. Final value is $+2$; reflecting after the first touch of -1 sends the final value to $-1-(2+1) = -4$, i.e. 3 ups and 7 downs: $\binom{10}{3} = 120$ bad orderings. Good: $210 - 120 = 90$, so $\mathbb{P} = \frac{90}{210} = \frac{3}{7} = \dfrac{a-b+1}{a+1}$ ✓.

Step 4. (c) If the first order is a sell (probability $\frac{4}{10}$) a tie is certain. If the first is a buy, ties correspond one-to-one (by reflection) with "first order is a sell" paths. So $\mathbb{P}(\text{tie}) = \dfrac{2b}{a+b} = \dfrac{8}{10} = \dfrac{4}{5}$. Consistency with (a): no tie $= \frac{1}{5}$.

Step 5. (d) Return times of the simple fair walk satisfy $f_{2n} = \dfrac{u_{2n}}{2n-1}$, where $u_{2n} = \binom{2n}{n}4^{-n}$. For $2n = 10$: $u_{10} = \frac{252}{1024}$, so $f_{10} = \frac{252}{1024 \cdot 9} = \frac{28}{1024} = \frac{7}{256} \approx 0.0273$.

Step 6. Cross-check via Catalan: first return at $2n$ means first step up, a Dyck-type path of length $2n-2$ staying strictly above 0, then a step down; times 2 for the sign. Count $= 2C_{n-1} = 2C_4 = 28$ paths, so $\frac{28}{1024}$ ✓.

Answer. (a) $\frac{1}{5}$ (b) $\frac{3}{7}$ (c) $\frac{4}{5}$ (d) $\frac{7}{256}$.

Why it works – the big idea

The ballot theorem is the reflection principle in disguise: "never touches the tie line" is "(all paths) − (paths that touch it) = (all paths) − (paths ending on the mirror side)". Catalan numbers are the special case with equal totals: $C_n = \frac{1}{n+1}\binom{2n}{n}$ counts the paths that never go below 0 and end at 0.

Analogy – make it intuitive

Two candidates running a long counting night. If the challenger gets the first vote, the favourite must catch up at some point, so a tie is certain. The mirror twin of "favourite leads, then gets caught" is "challenger leads first": equally many.

Common trap

"Never behind" ($\frac{a-b+1}{a+1}$) is *not* the same as "strictly ahead" ($\frac{a-b}{a+b}$). The first allows ties, the second does not. Check with $(3,2)$: $\frac{1}{2}$ vs $\frac{1}{5}$.

Part IV: Walls, cycles and higher dimensions

Problem 7 | Inventory with a wall

Difficulty: ⋆⋆⋆ **Target time:** 4 min **Topic:** reflecting barrier, unfolding

A market maker's inventory moves by ± 1 each tick with probability $\frac{1}{2}$, on positions $0, 1, \ldots, 12$. At 12 she is forced to liquidate (the process stops).

(a) At position 0 a hard wall pushes her to 1 with certainty. Starting from 4, what is the expected number of ticks until liquidation?

(b) Now the wall at 0 is "sticky": from 0 she stays at 0 w.p. $\frac{1}{2}$ and goes to 1 w.p. $\frac{1}{2}$. Starting from 0, what is the expected time to reach 12?

Step-by-step solution

Step 1. (a) *Unfold the wall.* If S_n is a fair walk on $\mathbb{Z}$, then $|S_n|$ is exactly the inventory process (from 0 it must go to 1; elsewhere it is fair). Reaching 12 means S_n hits $+12$ or -12 starting from 4.

Step 2. Shift to $[0, 24]$: start $k = 16$, $N = 24$, and $\mathbb{E}[T] = k(N-k) = 16 \cdot 8 = 128$.

Step 3. Check with the closed form $D_k = N^2 - k^2 = 144 - 16 = 128$, which satisfies $D_{k+1} - 2D_k + D_{k-1} = -2$, $D_0 = 1 + D_1$, $D_{12} = 0$ ✓.

Step 4. (b) With a sticky wall $D_0 = 1 + \frac{1}{2}D_0 + \frac{1}{2}D_1 \Rightarrow D_0 = 2 + D_1$. Take $D_k = -k^2 - k + c$: it satisfies the interior equations; the wall condition gives $-0 - 0 + c = 2 + (-1 - 1 + c) \Rightarrow$ true for all c, and $D_{12} = 0$ fixes $c = 12^2 + 12 = 156$.

Step 5. So $D_k = (N-k)(N+k+1)$, and from 0: $D_0 = 12 \cdot 13 = 156$ (and $D_1 = 11 \cdot 14 = 154$, indeed $D_0 = 2 + D_1$).

Answer. (a) 128 ticks (b) 156 ticks.

Why it works – the big idea

A reflecting wall does not change the *shape* of the walk, only the track it runs on. Replace the walker by its absolute value and the wall disappears, the track doubles in length, and you are back to a plain two-barrier fair walk. A sticky wall adds one extra tick on average each time you touch it, which costs N extra ticks in total from 0.

Analogy – make it intuitive

A ball in a corridor with a springy wall on the left and an open door on the right. Instead of watching it bounce, imagine a mirror-image corridor behind the wall: the ball simply continues straight into the ghost corridor, and the double-length corridor has two doors.

Problem 8 | The ant on a hexagon

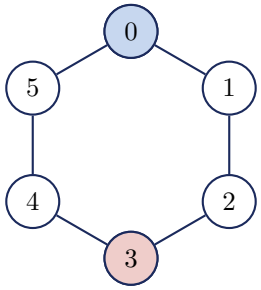
Difficulty: ⋆⋆⋆ **Target time:** 4 min **Topic:** cycles, cover time, last vertex

An ant sits at a vertex of a regular hexagon. Each second it moves to one of its two neighbours with probability $\frac{1}{2}$ each.

(a) What is the expected time to reach the opposite vertex?

(b) What is the expected time until it has visited *all* six vertices?

(c) What is the probability that the opposite vertex is the *last* one visited?



Step-by-step solution

Step 1. (a) Unwrap the cycle: reaching vertex 3 from 0 is the same as the unwrapped walk hitting $+3$ or -3. Fair ruin with $k = 3$, $N = 6$: $\mathbb{E}[T] = 3 \cdot 3 = 9$.

Step 2. (b) When j distinct vertices have been visited they form a contiguous arc, and the ant is at one end. It needs to step beyond either end: exit from an interval of length $j + 1$ starting at distance 1 from one end, which takes $1 \cdot j = j$ steps on average. Sum over $j = 1, \dots, 5$: $1 + 2 + 3 + 4 + 5 = 15$ (in general $\frac{N(N-1)}{2}$).

Step 3. (c) Let the target "last" vertex be vertex j at clockwise distance j from the start. For it to be last, the ant must visit both neighbours $j - 1$ and $j + 1$ first. The ant first reaches either $j - 1$ or $j + 1$ (never crossing j): by fair ruin on the long arc of length $N - 2$ it reaches $j - 1$ first w.p. $\frac{N-j-1}{N-2}$ and $j + 1$ first w.p. $\frac{j-1}{N-2}$. From the neighbour it reached, it must then walk the long way to the *other* neighbour before stepping onto j: starting at distance 1 from j on an interval of length $N - 1$, the probability is $\frac{1}{N-1}$.

Step 4. Total: $\dfrac{(N-j-1)+(j-1)}{(N-2)(N-1)} \cdot 1 = \dfrac{N-2}{(N-2)(N-1)} = \dfrac{1}{N-1}$, independent of j. For $N = 6$: $\dfrac{1}{5}$.

Answer. (a) 9 (b) 15 (c) $\frac{1}{5}$.

Why it works – the big idea

On a cycle, the last vertex visited is *uniform* over all the other $N - 1$ vertices. That is a famous surprise: the farthest vertex is no more likely to be last than the immediate neighbour. The reason is that to be last a vertex must be "walled in" on both sides, and the two competing effects cancel exactly.

Analogy – make it intuitive

A painter painting a circular fence one board at a time. The painted section only grows at its two edges, so he repaints the same boards a lot, and the board that happens to be last is determined by being "cut off from both sides", which is equally hard for every board.

Problem 9 | Asymmetric barriers with a biased coin

Difficulty: ⋆⋆⋆ **Target time:** 3 min **Topic:** ruin, duration, Wald consistency

A trader's P&L moves $+1$ with probability 0.6 and -1 with probability 0.4 each trade. She quits at $+4$ (target) or -2 (stop-out).

(a) What is the probability she hits the $+4$ target?

(b) What is the expected number of trades?

(c) What is her expected final P&L? Verify it using (b).

Step-by-step solution

Step 1. Shift by $+2$: start $k = 2$, $N = 6$. $r = \frac{0.4}{0.6} = \frac{2}{3}$.

Step 2. (a) $W_2 = \dfrac{1-r^2}{1-r^6} = \dfrac{1-\frac{4}{9}}{1-\frac{64}{729}} = \dfrac{5/9}{665/729} = \dfrac{5\cdot 81}{665} = \dfrac{81}{133} \approx 0.609.$

Step 3. (b) Wald: $\mathbb{E}[T] = \dfrac{NW_2 - k}{p-q} = \dfrac{6\cdot\frac{81}{133} - 2}{0.2} = \dfrac{\frac{486-266}{133}}{0.2} = \dfrac{220/133}{1/5} = \dfrac{1100}{133} \approx 8.27.$

Step 4. (c) Final P&L $= +4$ w.p. $\frac{81}{133}$, -2 w.p. $\frac{52}{133}$: $\mathbb{E} = \dfrac{324-104}{133} = \dfrac{220}{133} \approx 1.654.$

Step 5. Check: drift times duration $= 0.2 \cdot \frac{1100}{133} = \frac{220}{133}$ ✓.

Answer. (a) $\frac{81}{133} \approx 0.609$ (b) $\frac{1100}{133} \approx 8.27$ trades (c) $\frac{220}{133} \approx \$1.65$.

Why it works – the big idea

Optional stopping on the martingale $S_n - (p-q)n$ gives $\mathbb{E}[S_T] = (p-q)\mathbb{E}[T]$. The right side is "drift $\times$ time" and the left side is "average exit P&L". Any answer for $\mathbb{E}[T]$ should pass this check. It takes ten seconds and catches arithmetic slips.

Analogy – make it intuitive

Think of a conveyor belt moving you right at speed $p - q = 0.2$ per trade. Your average displacement at exit divided by the belt speed gives the average time on the belt.

Problem 10 | One dimension, two dimensions, three dimensions

Difficulty: ⋆⋆⋆⋆ **Target time:** 4 min **Topic:** return probabilities, recurrence

A drunk takes steps of length 1 at random on the integer lattice $\mathbb{Z}^d$, each of the $2d$ axis-neighbours equally likely.

(a) Find the probability of being at the origin after 4 steps for $d = 1, 2, 3$.

(b) Explain why the drunk returns to the origin with probability 1 for $d = 1, 2$ but not for $d = 3$.

(c) For $d = 3$ the return probability is about 0.34. What is the expected number of visits to the origin (counting the start)?

Step-by-step solution

Step 1. (a) $d = 1$: $\binom{4}{2}/2^4 = \frac{6}{16} = \frac{3}{8}$.

Step 2. $d = 2$: rotate the lattice by 45°. The step set $\{N, S, E, W\}$ becomes $\{(\pm 1, \pm 1)\}$, so the two coordinates are *independent* fair 1D walks: $\left(\frac{3}{8}\right)^2 = \frac{9}{64}$. (Direct count: closed 4-step

walks $= \binom{4}{2}^2 = 36$ out of $4^4 = 256$.)

Step 3. $d = 3$: count closed 4-step walks. Either use one axis twice in each direction: 3 axes $\times\binom{4}{2} = 18$; or use two different axes once in each direction: 3 axis-pairs $\times 4! = 72$. Total 90 out of $6^4 = 1296$: $\dfrac{90}{1296} = \dfrac{5}{72} \approx 0.0694$.

Step 4. (b) The probability of being at the origin after $2n$ steps is about $(\pi n)^{-d/2} \times$const. The expected number of visits is the sum over n. For $d = 1$: $\sum n^{-1/2} = \infty$; for $d = 2$: $\sum n^{-1} = \infty$; for $d = 3$: $\sum n^{-3/2} < \infty$. The number of returns is geometric: if the return probability were $P < 1$, the mean number of returns would be $\frac{P}{1-P} < \infty$. An infinite mean forces $P = 1$ (recurrent). A finite mean means $P < 1$ (transient).

Step 5. (c) With return probability $P \approx 0.34$, the number of visits (counting the start) is geometric with success probability $1 - P$, so the mean is $\dfrac{1}{1-P} \approx \dfrac{1}{0.66} \approx 1.5$.

Answer. (a) $\frac{3}{8}$, $\frac{9}{64}$, $\frac{5}{72}$ (b) divergent series $\Rightarrow$ recurrent for $d = 1, 2$; convergent for $d = 3 \Rightarrow$ transient (c) ≈ 1.5 visits.

Why it works – the big idea

Recurrence is decided by one integral: whether $\sum_n \mathbb{P}(\text{at origin at time } 2n)$ diverges. Lower dimensions have so little room that the walker keeps bumping into the origin; from dimension 3 on there are enough "escape routes" that, with positive probability, the walker wanders off forever.

Analogy – make it intuitive

Kakutani's quip: *a drunk man will eventually find his way home, but a drunk bird may get lost forever.* The man walks on a street grid (2D), the bird flies in 3D.

Answer key and final Stage-1 checklist

Answer key

#	Answer
1	(a) $\frac{1}{3}$; (b) 50 s; (c) \$0; (d) $-\$0.50$
2	(a) $\frac{8}{9}$; (b) 7; (c) $\frac{1}{3}$
3	(a) 21; (b) $\frac{91}{3}$; (c) 17
4	(a) $\frac{8}{27}$; (b) $\mathbb{E}[M] = 2$, $\mathrm{Var}(M) = 6$; (c) $\frac{8}{27}$
5	(a) $\frac{401930}{2^{20}} \approx 0.3833$; (b) $\frac{21700}{2^{20}} \approx 0.0207$; (c) $\frac{125970}{2^{20}} \approx 0.1201$
6	(a) $\frac{1}{5}$; (b) $\frac{3}{7}$; (c) $\frac{4}{5}$; (d) $\frac{7}{256}$
7	(a) 128; (b) 156
8	(a) 9; (b) 15; (c) $\frac{1}{5}$
9	(a) $\frac{81}{133}$; (b) $\frac{1100}{133} \approx 8.27$; (c) $\frac{220}{133} \approx 1.654$
10	(a) $\frac{3}{8}$, $\frac{9}{64}$, $\frac{5}{72}$; (b) $\sum$ diverges for $d \le 2$, converges for $d = 3$; (c) ≈ 1.5

Stage-1 checklist for any random-walk question

1. **Identify the state space:** where are the absorbing, reflecting and free states?
2. **Fair or biased?** Fair: linear probabilities and quadratic times. Biased: geometric with $r = q/p$.
3. **Shortcut hunt (10 seconds):** halfway trick, martingale/Wald, unfolding a wall, ties as pauses, mirror trick.
4. **Parity and off-by-one:** walks of n steps only reach levels with the same parity as n.
5. **Sanity check:** probabilities in $[0, 1]$; fair games keep the expected fortune; weighted conditional times add to the unconditional time; drift $\times$ time = expected displacement.